Advanced Causal Inference

Lecture 15: DiD/SC/RD

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This lecture note is partially based on the lecture notes/textbook of Joshua Shea

Outline

1. Overview
2. Difference-in-Differences/Two-way Fixed Effects Models
3. Synthetic Control
4. Regression Discontinuity
5. Additional Methods
6. Assignment

Overview

- We will consider causal inference methods that are commonly used in social sciences

Outline

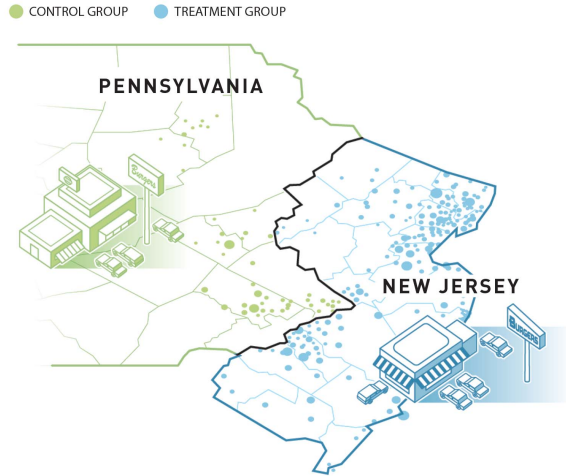
1. Overview
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Panel Data

- Consider a panel dataset with N units observed at two time points:
  - At time $t = 0$, all N units are under control;
  - Between time $t = 0$ and $t = 1$, some units receive treatment
- Y_{it} = outcome of unit i at time $t \in \{0, 1\}$
- A_i = indicator of whether unit i receives treatment at time $t = 1$
- $Y_{it}(a)$ = potential outcome of unit i at time $t \in \{0, 1\}$, had it received $A_i = a$ between time 0 and 1
- X_i = pre-treatment covariates measured before time $t = 0$

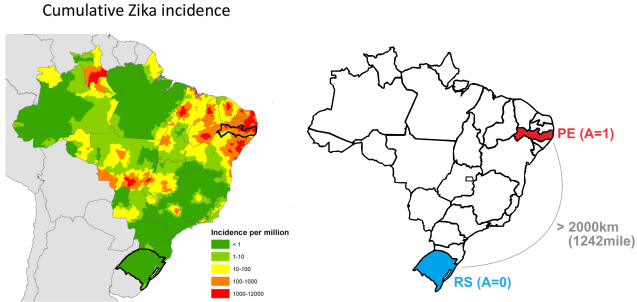
Example: Minimum Wage (Card and Krueger, 1994)

- Intervention:
  1992 minimum wage increase in NJ
- Time $t = 0, 1$:
  Feb 1992, Dec 1992
- Study units N:
  358 fast food restaurants in NJ and PA
- Outcome Y_{it}:
  FTE (full-time equivalent; FTE=0.5 means half-time worker)
- $A_i = 1$: NJ, 291; $A_i = 0$: Eastern PA, 67
- Covariates X:
  Fast-food franchises (Burger King, KFC, Roy's, Wendy's)
- Hypothesis:
  The mimimum wage increase resulted in a drop in FTE positions



Example: 2015 Zika Virus Outbreak in Brazil (Park et al., 2024)

- Intervention:
  2015 Zika virus outbreak in Brazil
- Time $t = 0, 1$:
  2014 and 2016
- Study units N:
  673 municipalities in Brazil
- Outcome Y_{it}:
  birth rate (number of live human births per 1000 persons)



Brady et al. (2019) PLOS Medicine

- $A_i = 1$: Pernambuco (PE), 185; $A_i = 0$: Rio Grande do Sul (RS), 488
- Covariates X:
  log population, population density, proportion of females in 2014
- Hypothesis:
  The Zika virus outbreak resulted in a drop in birth rate

Estimand: ATT

- When the intervention is an external shock, we are typically interested in its effect on the units that were exposed to it

$$\tau_{\text{ATT}} = \mathrm{E}\big\{Y_{i1}(1) - Y_{i1}(0) \,\big|\, A_i = 1\big\}$$

- In the previous examples:

$\tau_{\text{ATT}} =$ effect of the minimum wage increase on FTE in NJ fast food restaurants

$\tau_{\text{ATT}} =$ effect of the Zika virus outbreak on birth rate in PE

Assumption

Assumption (Standard Panel Data)

(SUTVA) $Y_{it} = Y_{it}(A_i)$

(Positivity) $\Pr(A_i = 1) > 0$ *and* $e(X_i) := \Pr(A_i = 1|X_i) < 1$

- Under the assumption,

$$\tau_{\text{ATT}} = \mathrm{E}\big\{Y_{i1}(1) - Y_{i1}(0) \,\big|\, A_i = 1\big\} = \mathrm{E}\big(Y_{i1} \,\big|\, A_i = 1\big) - \mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1\big\}$$

- The second piece $\mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1\big\}$ is counterfactual

Selection Only on Observables

- Under the assumption,

$$\tau_{\text{ATT}} = \mathrm{E}\big\{Y_{i1}(1) - Y_{i1}(0) \,\big|\, A_i = 1\big\} = \mathrm{E}\big(Y_{i1} \,\big|\, A_i = 1\big) - \mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1\big\}$$

- If A_i was randomized,

$$\mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1\big\} = \mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 0\big\} = \mathrm{E}\big(Y_{i1} \,\big|\, A_i = 0\big)$$

- τ_{ATT} can be estimated from the 2-sample t-test or linear regression

- Selection only on observables = no unmeasured confounder = $Y_{it}(a) \perp\!\!\!\perp A_i | X_i$, then

$$\mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1\big\} = \mathrm{E}\left\{\frac{A_i Y_{i1}(0)}{\Pr(A_i = 1)}\right\} = \mathrm{E}\left[\frac{\mathrm{E}(A_i|X_i)\mathrm{E}\big\{Y_{i1}(0) \,\big|\, X_i\big\}}{\Pr(A_i = 1)}\right] = \mathrm{E}\left\{\frac{e(X_i)\mathrm{E}\big(Y_{i1} \,\big|\, A_i = 0, X_i\big)}{\Pr(A_i = 1)}\right\}$$

- If one assumes effect homogeneity, one can fit

$$Y_{i1} = \beta_0 + \tau_{\text{ATT}} A_i + \beta_X^{\intercal} X_i + \epsilon_i$$

and conduct a hypothesis test $H_0 : \tau_{\text{ATT}} = 0$

ATT Estimates Under Selection on Observables

- NJ Minimum Wage

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.27353    1.18188   6.154 2.05e-09 ***
A            0.55383    1.03573   0.535  0.59318
X1          -2.26018    1.19704  -1.888  0.05983 .
X2           2.67378    1.02428   2.610  0.00943 **
X3          -0.01014    1.38697  -0.007  0.99417
```

- No effect of NJ minimum wage increase on FTE

- Brazil Zika

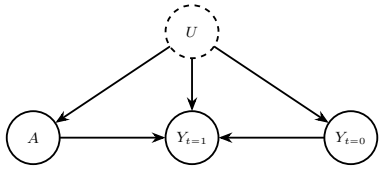
```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -5.6488     4.5127  -1.252    0.211
A             2.0807     0.2441   8.523  < 2e-16 ***
X1            0.8324     0.1175   7.087 3.49e-12 ***
X2            0.1449     0.1027   1.411    0.159
X3           16.3131     9.9453   1.640    0.101
```

- The Zika virus outbreak resulted in an increased birth rate in PE (?)

Selection on Unobservables

- In many panel data setting, $Y_{it}(a) \perp\!\!\!\perp A_i | X_i$ is not plausible

- In the previous examples, U = region-specific characteristics
  - In Brazil, the northeastern region has lower income levels and a climate favorable to mosquitoes,
    whereas the southern region has higher income levels and a climate less conducive to mosquitoes.



- This represents selection based on unobserved factors

- Although this is not a formal statistical test,
  one can heuristically assess whether selection on unobservables exists using the pre-treatment outcome Y_{i0}

No Anticipation

Assumption (No Anticipation)

$Y_{i0}(0) = Y_{i0}(1)$

- Regardless of whether a unit receives treatment at $t = 1$, the potential outcome at the baseline period $t = 0$ remains the same
- A future intervention does not affect past outcomes
- Plausible if the intervention is truly a "shock":
  - Brazil Zika: No one would have adjusted their family planning in 2014 in anticipation of a Zika outbreak in 2015
- May not be plausible if the intervention is anticipated:
  - Relocation of Indonesia's capital city: Although the decision to move the capital is made, the actual relocation takes time. As a result, outcomes prior to the actual move may be influenced by anticipation
- Under SUTVA and no anticipation, we have

$$Y_{i0} = Y_{i0}(A_i) = Y_{i0}(0) = Y_{i0}(1)$$

which means there is no individual-level treatment effect of A_i on Y_{i0}

A Heuristic Approach for Assessing Selection on Unobservables

- If there the ignorability holds, i.e., $Y_{it}(a) \perp\!\!\!\perp A|X$, then

$$\begin{aligned} 0 &= \mathrm{E}\{Y_{i0}(1) - Y_{i0}(0)|X\} \\ &= \mathrm{E}\{Y_{i0}(1) \,|\, A = 1, X\} - \mathrm{E}\{Y_{i0}(0) \,|\, A = 0, X\} \\ &= \mathrm{E}(Y_{i0} \,|\, A = 1, X) - \mathrm{E}(Y_{i0} \,|\, A = 0, X) \end{aligned}$$

- Therefore, one can consider a linear regression model:

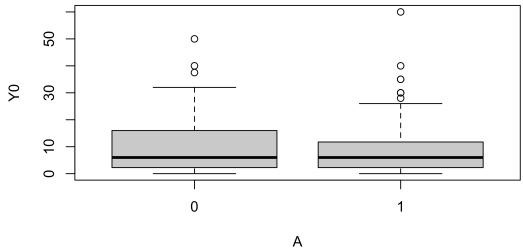
$$\texttt{lm}(Y_{i0} \sim 1 + A_i + X_i)$$

and conduct a hypothesis test $H_0 : \beta_A = 0$

A Heuristic Approach for Assessing Selection on Unobservables

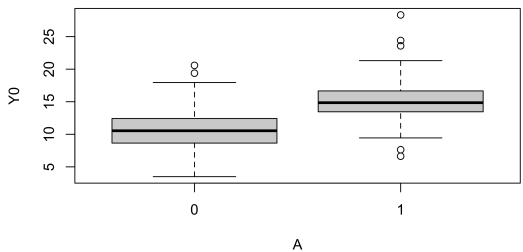
- NJ Minimum Wage

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.8197     1.3042   8.296  2.3e-15 ***
A            -2.2690     1.1429  -1.985 0.047892 *
X1           -4.6866     1.3209  -3.548 0.000441 ***
X2            1.0275     1.1303   0.909 0.363921
X3            0.4175     1.5305   0.273 0.785153
```

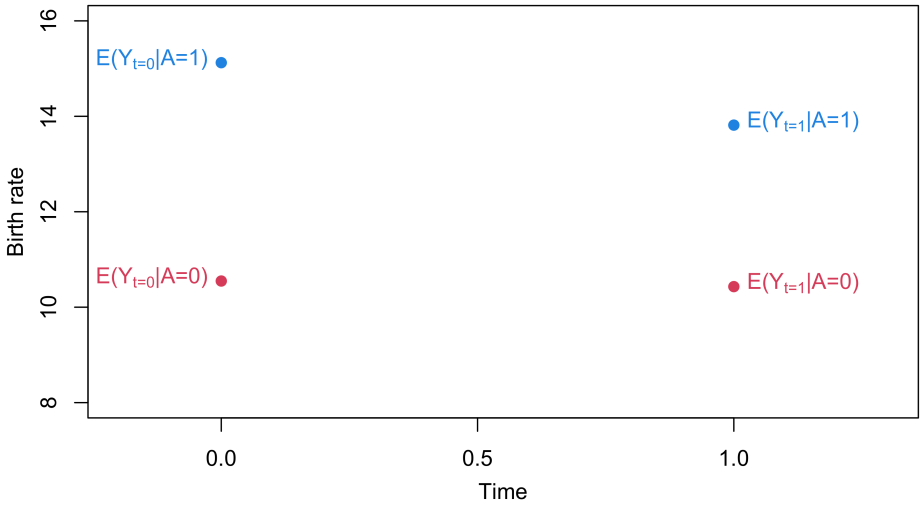


- Brazil Zika

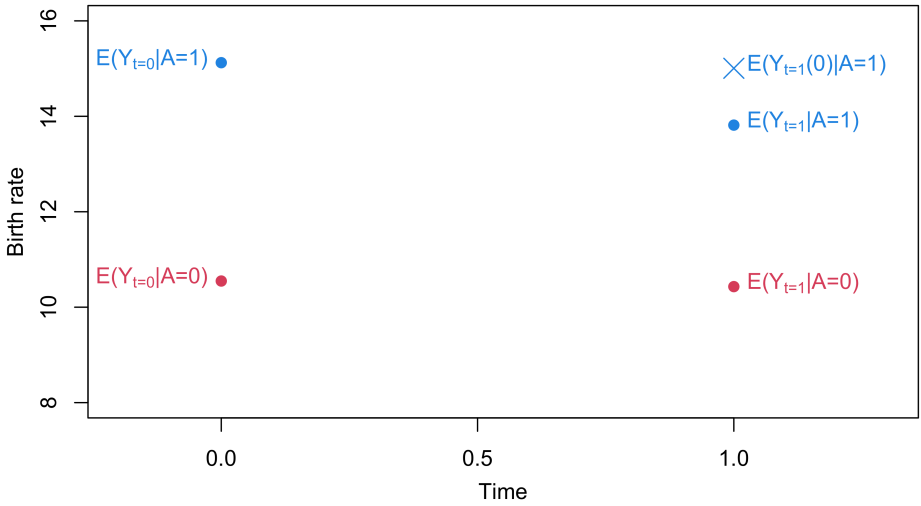
```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.09113    4.37078  -1.165   0.2445
A            3.20471    0.23620  13.568  < 2e-16 ***
X1           0.88308    0.11410   7.740  3.7e-14 ***
X2           0.17601    0.09927   1.773   0.0767 .
X3          14.31199    9.63896   1.485   0.1381
```



Visual Summary of the Zika Virus Example



Visual Summary of the Zika Virus Example



Parallel Trends

Assumption (Parallel Trends)

$$\mathrm{E}\{Y_{i1}(0) - Y_{i0}(0) | A = 1, X\} = \mathrm{E}\{Y_{i1}(0) - Y_{i0}(0) | A = 0, X\}$$

- The change of $Y_t(0)$ over time is the same in the treated and control groups

- The PT assumption is equao to the **additive equi-confounding**: (Sofer et al., 2016)

$$\mathrm{E}\{Y_{i1}(0) | A = 1, X\} - \mathrm{E}\{Y_{i1}(0) | A = 0, X\} = \mathrm{E}\{Y_{i0}(0) | A = 1, X\} - \mathrm{E}\{Y_{i0}(0) | A = 0, X\}$$

- The degree of confounding between $A - Y$ does not change over time

- Under SUTVA, no anticipation, PT, we have

$$\begin{aligned}
\mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1, X\big\} &= \mathrm{E}\big\{Y_{i0}(0) \,\big|\, A_i = 1, X\big\} + \mathrm{E}\big\{Y_{i1}(0) - Y_{i0}(0) \,\big|\, A_i = 0, X\big\} \\
&= \mathrm{E}\big\{Y_{i0}(1) \,\big|\, A_i = 1, X\big\} + \mathrm{E}\big\{Y_{i1}(0) - Y_{i0}(0) \,\big|\, A_i = 0, X\big\} \\
&= \mathrm{E}\big(Y_{i0} \,\big|\, A_i = 1, X\big) + \mathrm{E}\big(Y_{i1} - Y_{i0} \,\big|\, A_i = 0, X\big)
\end{aligned}$$

Standard Difference-in-Differences

Assumption (Standard DiD Assumptions)

(SUTVA)	$Y_{it} = Y_{it}(A_i)$
(Positivity)	$\Pr(A_i = 1) > 0$ *and* $e(X_i) := \Pr(A_i = 1 \mid X_i) < 1$
(No anticipation)	$Y_{i0}(0) = Y_{i0}(1)$
(Parallel trends)	$\mathrm{E}\{Y_{i1}(0) - Y_{i0}(0) \mid A = 1, X\} = \mathrm{E}\{Y_{i1}(0) - Y_{i0}(0) \mid A = 0, X\}$

- Under the standard DiD assumptions, the ATT can be identified:

$$\begin{aligned}
\tau_{\mathsf{ATT}} &= \mathrm{E}\big\{Y_{i1}(1) - Y_{i1}(0) \,\big|\, A_i = 1\big\} \\
&= \mathrm{E}\big(Y_{i1} \,\big|\, A_i = 1\big) - \mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1\big\} \\
&= \mathrm{E}\big(Y_{i1} \,\big|\, A_i = 1\big) - \mathrm{E}_X\big[\mathrm{E}\big\{Y_{i1}(0) \,\big|\, A_i = 1, X\big\} \,\big|\, A_i = 1\big] \\
&= \mathrm{E}\big(Y_{i1} - Y_{i0} \,\big|\, A_i = 1\big) - \mathrm{E}_X\big\{\mathrm{E}\big(Y_{i1} - Y_{i0} \,\big|\, A_i = 0, X\big) \,\big|\, A_i = 1\big\}
\end{aligned}$$

- If no covariate, we have $\tau_{\mathsf{ATT}} = \mathrm{E}\big(Y_{i1} - Y_{i0} \,\big|\, A_i = 1\big) - \mathrm{E}\big(Y_{i1} - Y_{i0} \,\big|\, A_i = 0\big)$, justifying the DiD form

Additional Discussions

- The concept of DiD dates back to the mid-19th century (Snow, 1856)

- The standard DiD model can be falsified
  - Suppose Y is binary
  - Under the standard DiD model, we have $\mathrm{E}\{Y_{i1}(0) \,|\, A_i = 1, X\} = \mathrm{E}(Y_{i0} \,|\, A_i = 1, X) + \mathrm{E}(Y_{i1} - Y_{i0} \,|\, A_i = 0, X)$
  - However, the right-hand side need not lie within the unit interval

- The EIF for the ATT in the nonparametric model is given by Sant'Anna and Zhao (2020)

- Callaway and Sant'Anna (2021): Extension to multiple periods and multiple treatments

- Pre-trend check: Using multiple pre-treatment periods,

$$H_{0,\text{pre}}: \ \mathrm{E}(Y_{it} - Y_{i,t-1} | A_i = 1) = \mathrm{E}(Y_{it} - Y_{i,t-1} | A_i = 0) \ , \quad t \in \{-m, \ldots, 0\}$$

  - If H_0 is rejected, this suggests that the PT assumption is less likely to hold
  - If H_0 is not rejected, this suggests that the PT assumption is more likely to hold
  - However, these conclusions are only indicative, not definitive:
    the PT assumption may still hold even if $H_{0,\text{pre}}$ is rejected, and it may fail even if $H_{0,\text{pre}}$ is not rejected.

Two-way Fixed Effects Models

- A canonical two-way fixed effects model (TWFE) for $t \in \{0, 1\}$:

$$Y_{it} = \alpha_i + \beta_t + \tau t A_i + \gamma^\intercal X_i + \epsilon_{it}$$

  - α_i = unit fixed effect
  - β_t = time fixed effect
  - τ = causal effect of treatment
  - A_i = $\mathbb{1}\{\text{Unit } i \text{ receives treatment at time } 1\}$
  - X_i = covariates
  - ϵ_{it} = error term

- Note that α_i and β_t are modeled as fixed effects (dummy variables)
  - Widely used in the social sciences
  - In statistics, random effects are often preferred

- The OLS estimator of τ is reported along with its SE
  - Residuals ϵ_{it} are usually correlated within units over time, so SEs are often clustered at the unit level i

Event Study Analysis

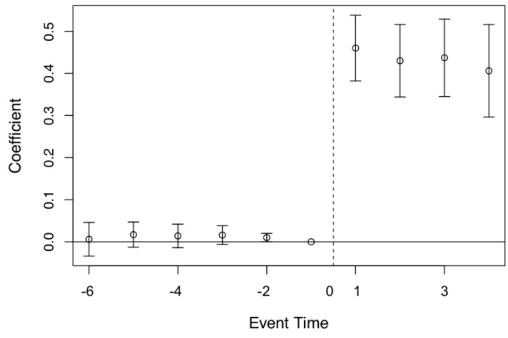
- Consider multiple time periods:

$$Y_{it} = \alpha_i + \beta_t + \sum_{k=-m}^{M} \tau_k \mathbb{1}(t = k) A_i + \gamma^\intercal X_i + \epsilon_{it}$$

$$= \begin{cases} & & \vdots & & & \\ \alpha_i & +\beta_{-1} & +\tau_{-1} A_i & +\gamma^\intercal X_i & +\epsilon_{i,-1} & \text{if } t = -1 \\ \alpha_i & +\beta_0 & & +\gamma^\intercal X_i & +\epsilon_{i0} & \text{if } t = 0 \\ \alpha_i & +\beta_1 & +\tau_1 A_i & +\gamma^\intercal X_i & +\epsilon_{i1} & \text{if } t = 1 \\ & & \vdots & & & \end{cases}$$

  - τ_k = causal effect at time $k \in \{-m, \ldots, M\}$

- This event study is introduced to detect both immediate and dynamic effects



https://mixtape.scunning.com/09-difference_in_differences

DiD vs. TWFE

- TWFE estimate is valid only under strong assumptions; say
  - No dynamic effects: treatment effect not change over time;
  - Unit-level effect homogeneity: treatment effect must be the same for all units;
  - Linear model: the linear model must be the true DGP

- Goodman-Bacon (2021): With multiple groups and staggered timing, TWFE estimate is a weighted average of many 2×2 DiD estimates

- If treatment effects are often heterogeneous, TWFE estimates can be biased, especially under staggered adoption or when treatment effects vary dynamically over time

- Using TWFE requires justifying the underlying assumptions;
  we must carefully argue why the key identification assumptions hold in the context of the study

Example: Minimum Wage

- Use `did` package; details are here https://bcallaway11.github.io/did/articles/did-basics.html

```
library(did)
dataset <- read.csv("https://libguides.princeton.edu/ld.php?content_id=68881201")

Y1 <- dataset$fullAfter
Y0 <- dataset$fullBefore
A <- as.numeric(dataset$location!="PA")
X1 <- as.numeric(dataset$chain=="kfc")
X2 <- as.numeric(dataset$chain=="burgerking")
X3 <- as.numeric(dataset$chain=="wendys")
N <- length(Y1)

data.CK <- data.frame(Y=c(Y0,Y1), Time=rep(c(0,1),each=N), A=c(A,A),
                      X1=c(X1,X1), X2=c(X2,X2), X3=c(X3,X3), ID=c(1:N,1:N))

ATT.CK <- att_gt(yname="Y",
                 tname="Time",
                 idname="ID",
                 gname="A",
                 xformla=~X1+X2+X3,
                 data=data.CK)
```

Example: Minimum Wage

```
Call:
att_gt(yname = "Y", tname = "Time", idname = "ID", gname = "A",
    xformla = ~X1 + X2 + X3, data = data.CK)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multiple Time Periods."
 Journal of Econometrics, Vol. 225, No. 2, pp. 200-230, 2021.
 <https://doi.org/10.1016/j.jeconom.2020.12.001>, <https://arxiv.org/abs/1803.09015>

Group-Time Average Treatment Effects:
 Group Time ATT(g,t) Std. Error [95% Pointwise  Conf. Band]
     1    1   2.4585      1.502         -0.5559      5.4728
---
Signif. codes: '*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Doubly Robust
```

Example: Minimum Wage

```
library(DRDID)

dataset <- read.csv("https://libguides.princeton.edu/ld.php?content_id=68881201")
Y1 <- dataset$fullAfter
Y0 <- dataset$fullBefore
A <- as.numeric(dataset$location!="PA")
X1 <- as.numeric(dataset$chain=="kfc")
X2 <- as.numeric(dataset$chain=="burgerking")
X3 <- as.numeric(dataset$chain=="wendys")
N <- length(Y1)

twfe_did_panel( y1=Y1, y0=Y0, D=A, covariates=cbind(X1,X2,X3))
```

- $\widehat{\tau}_{\text{TWFE}} = 2.93$; SE $= 1.76$; 95% CI $=$ (-0.51, 6.37)

Example: County Teen Employment Dataset

- Effect of Minimum Wage Policy on Teen Employment

- `did::mpdta`

t	year	the year of the observation
i	countyreal	a unique identifier for a particular county
X	lpop	the log of 1000s of population for the county
Y	lemp	the log of teen employment in the county
G	first.treat	the year that the state where the county is located raised its minimum wage it is set equal to 0 for counties that have minimum wages equal to the federal minimum wage over the entire period
A	treat	whether or not a particular county is treated in that year

Example: County Teen Employment Dataset

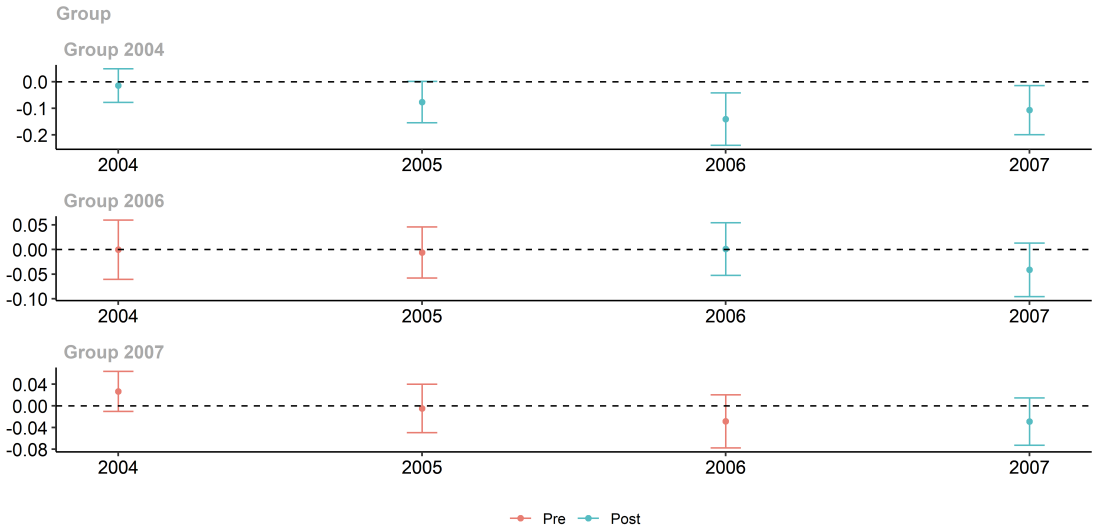
```
att_gt(yname="lemp", tname="year", idname="countyreal", gname="first.treat", xformla=~lpop, data=mpdta)

Group-Time Average Treatment Effects:
 Group Time ATT(g,t) Std. Error [95% Simult.  Conf. Band]
  2004 2004  -0.0145     0.0244       -0.0788      0.0497
  2004 2005  -0.0764     0.0299       -0.1552      0.0024
  2004 2006  -0.1404     0.0366       -0.2368     -0.0441 *
  2004 2007  -0.1069     0.0338       -0.1959     -0.0179 *
  2006 2004  -0.0005     0.0219       -0.0583      0.0573
  2006 2005  -0.0062     0.0183       -0.0544      0.0420
  2006 2006   0.0010     0.0190       -0.0492      0.0511
  2006 2007  -0.0413     0.0202       -0.0946      0.0121
  2007 2004   0.0267     0.0148       -0.0122      0.0656
  2007 2005  -0.0046     0.0156       -0.0456      0.0365
  2007 2006  -0.0284     0.0188       -0.0780      0.0211
  2007 2007  -0.0288     0.0169       -0.0732      0.0156
---
Signif. codes: '*' confidence band does not cover 0

P-value for pre-test of parallel trends assumption:  0.23267
Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Doubly Robust
```

Example: County Teen Employment Dataset

```
ggdid(att_gt(yname="lemp", tname="year", idname="countyreal", gname="first.treat", xformla=~lpop, data=mpdta))
```



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Synthetic Control Setting

- Consider a panel dataset with $N+1$ units observed over T time periods
  - For $t \in \{1, \ldots, T_0\}$, all $N+1$ units are in the control condition
  - Starting at $t = T_0 + 1$, only unit $i = 0$ receives the treatment, while the remaining units $i = 1, \ldots, N$ remain untreated

- Y_t = outcome of unit $i = 0$ at time $t \in \{1, \ldots, T\}$
- W_{it} = outcome of unit $i \in \{1, \ldots, N\}$ at time $t \in \{1, \ldots, T\}$

- $A_t = \mathbb{1}(T_0 < t)$; indicator function equal to 1 if t is in the post-treatment period, and 0 otherwise

- $Y_t(a)$ = potential outcome of unit $i = 0$ at time t, had it received $\cdots$
- $W_{it}(a)$ = potential outcome of unit i at time t, had it received $\cdots$

Synthetic Control Approach

- California tobacco control program in Abadie et al. (2010)

Treatment:	Proposition 99 (a large-scale tobacco control program) in 1988
Pre- and post-intervention periods:	$T_0 = 19$ (1970-1988) and $T_1 = 12$ (1989-2000) years
Outcome Y_t:	Per-capita cigarette sales of California in year t
Outcome W_{it}:	Per-capita cigarette sales of other 38 states in year t

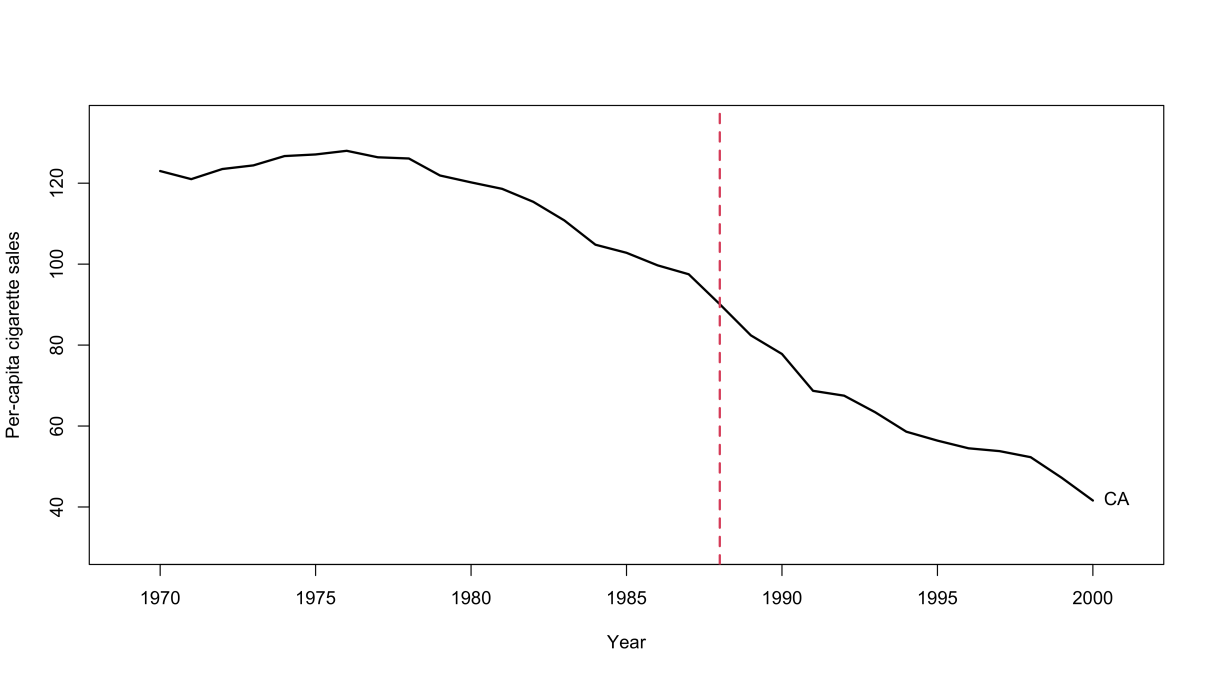
Synthetic Control Approach

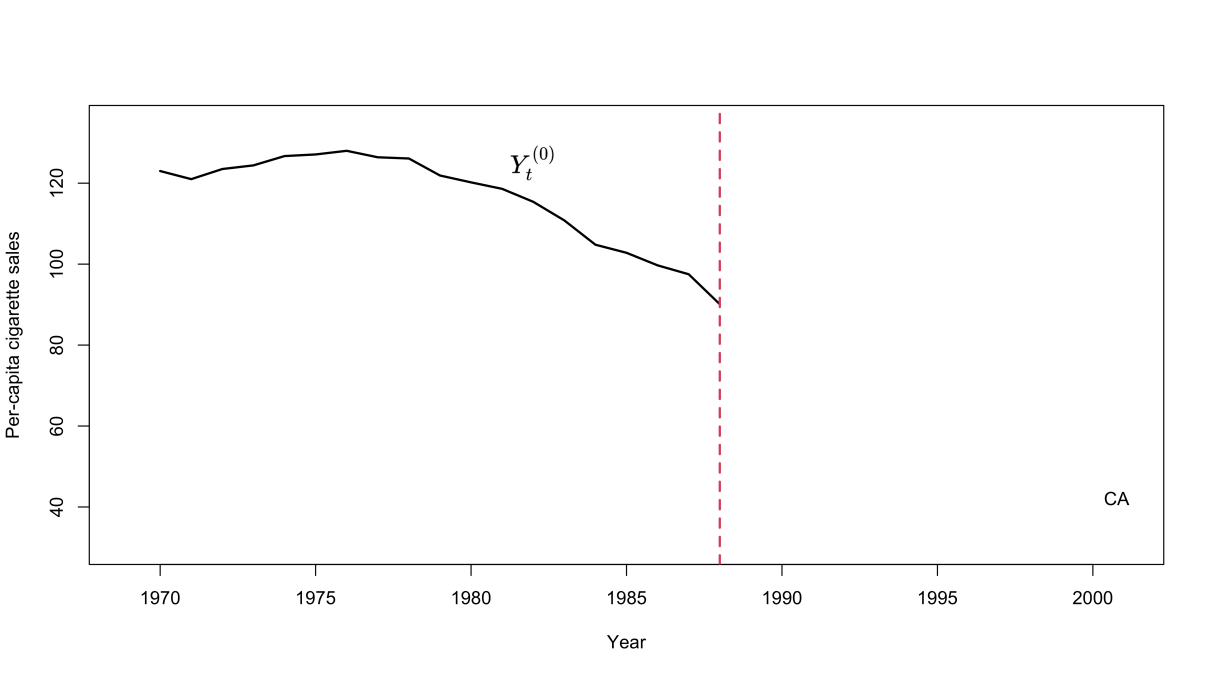
- California tobacco control program in Abadie et al. (2010)

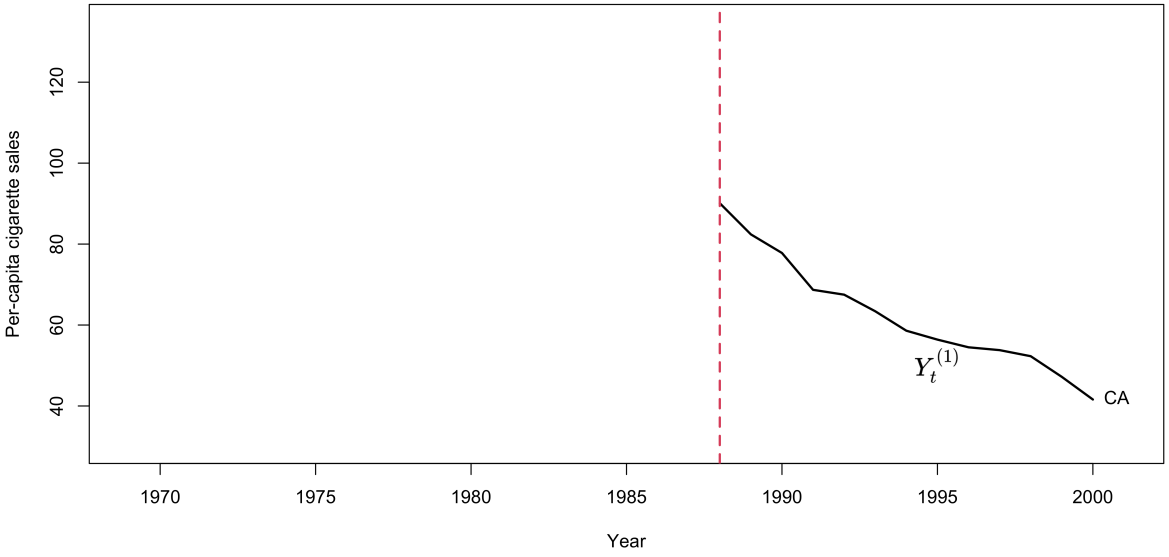
Treatment:	Proposition 99 (a large-scale tobacco control program) in 1988
Pre- and post-intervention periods:	$T_0 = 19$ (1970-1988) and $T_1 = 12$ (1989-2000) years
Outcome Y_t:	Per-capita cigarette sales of California in year t
Outcome W_{it}:	Per-capita cigarette sales of other 38 states in year t

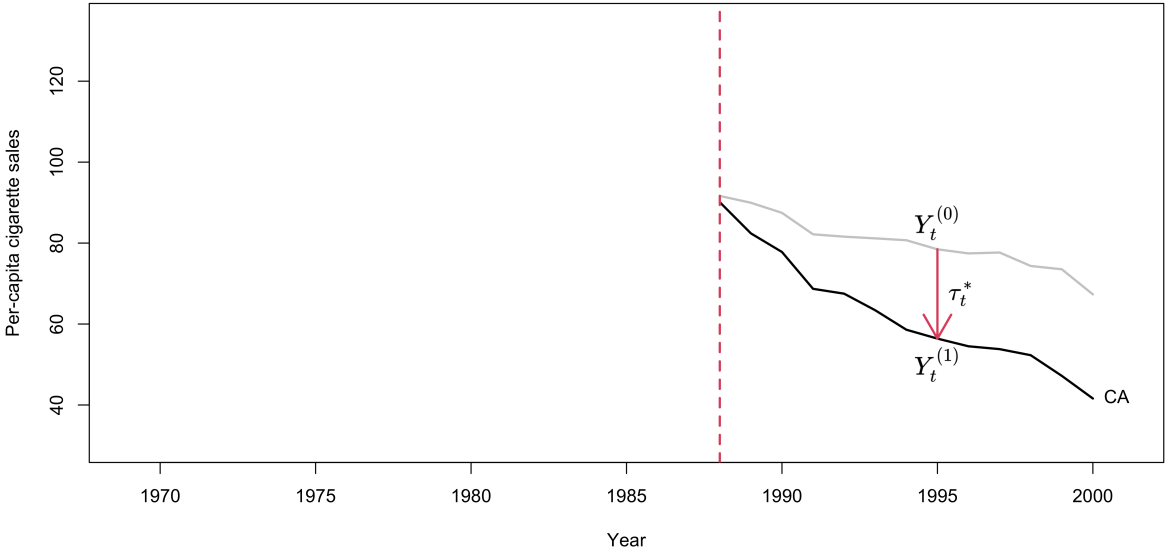
- Estimand: Effect of California's Proposition 99 on California's per-capita cigarette sales

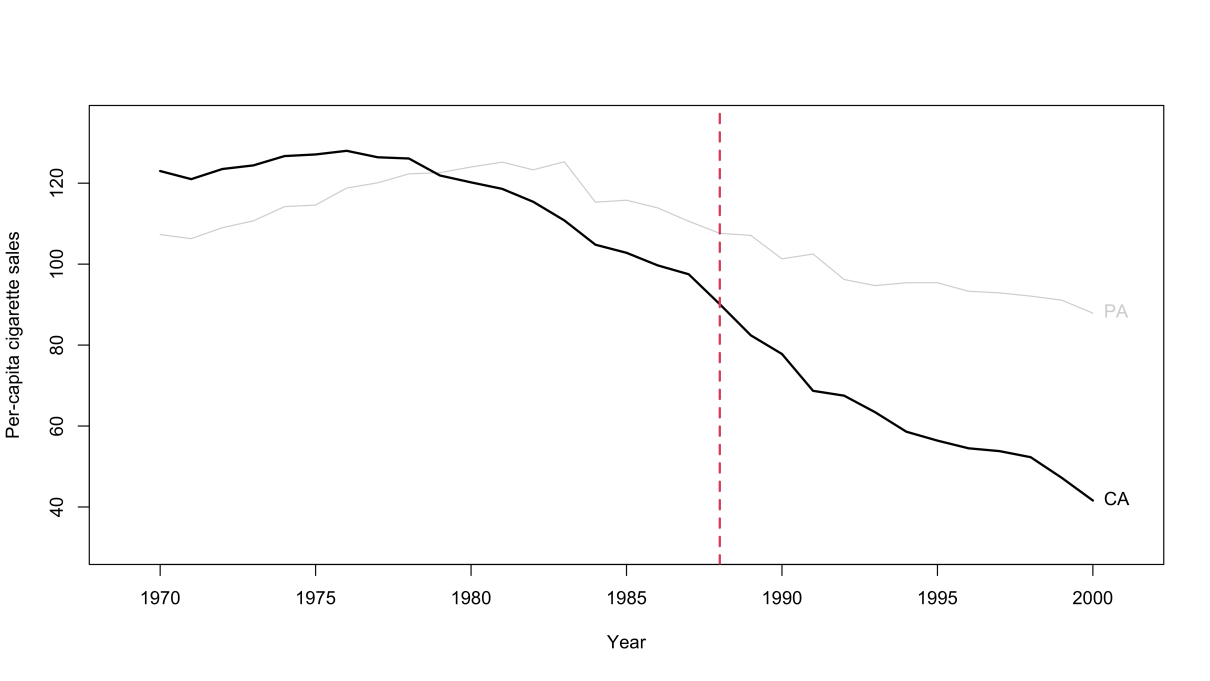
$$\tau_t = \mathrm{E}\big\{Y_t(1) - Y_t(0)\big\}\ , \quad 1988 < t$$

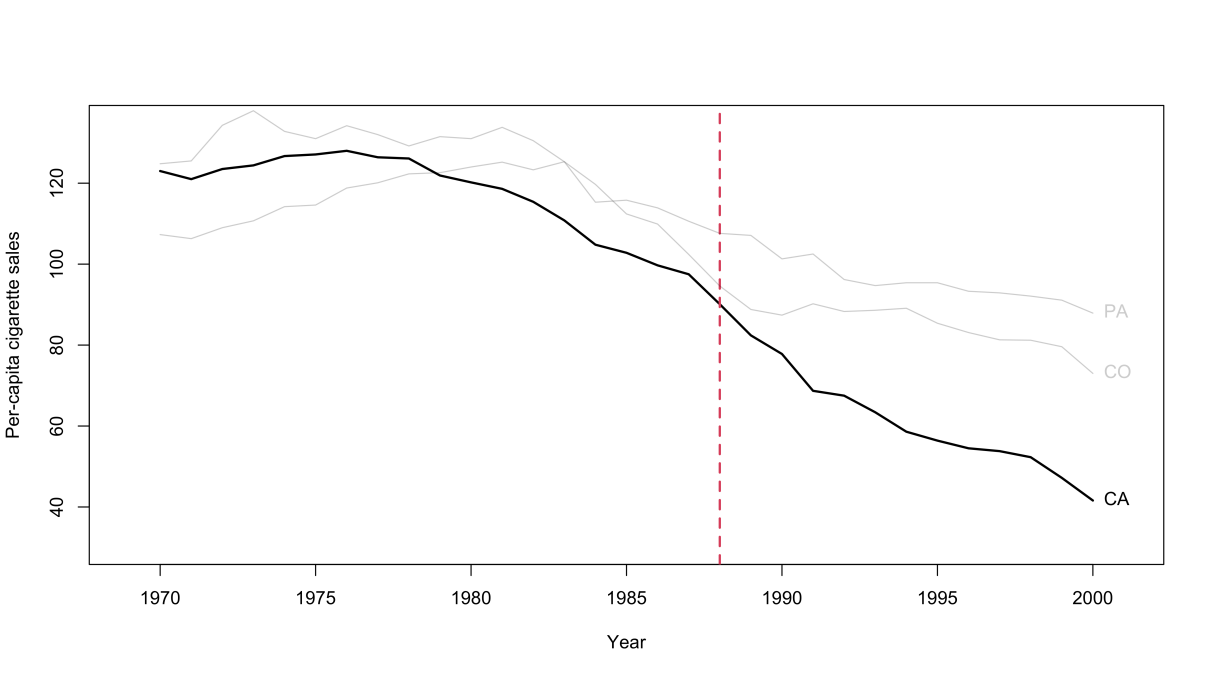

Per-capita cigarette sales
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120
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1990
1995
2000
Year
CA

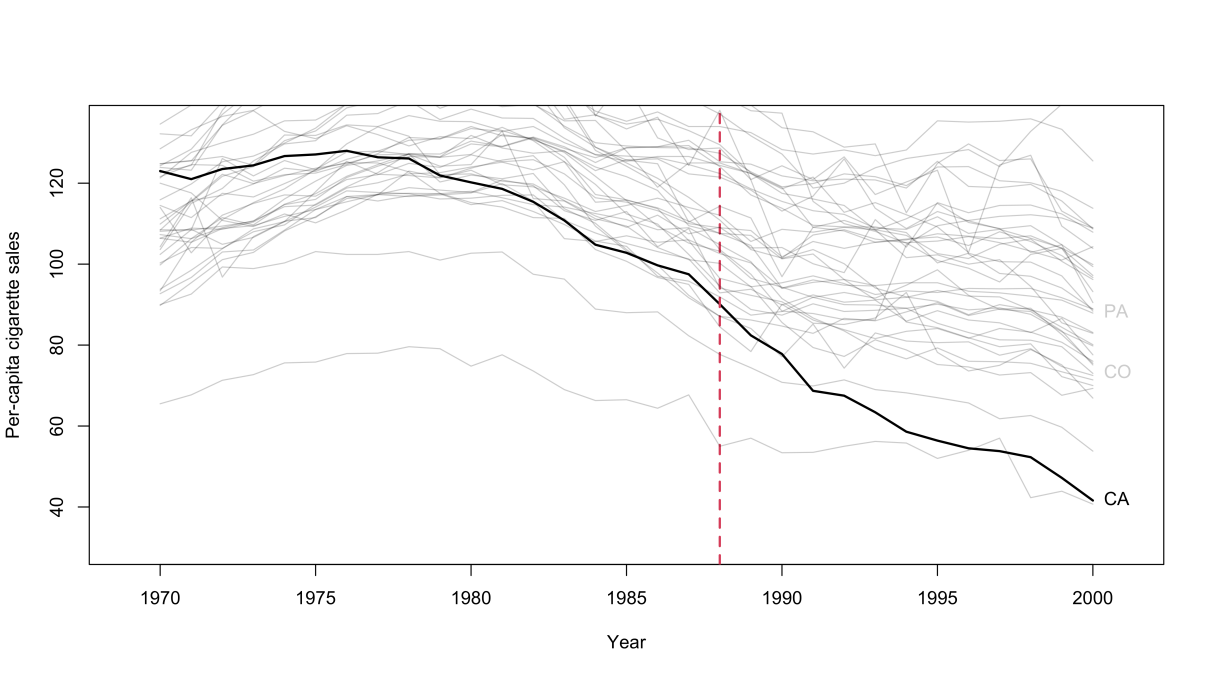

$Y_t^{(0)}$
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Per-capita cigarette sales
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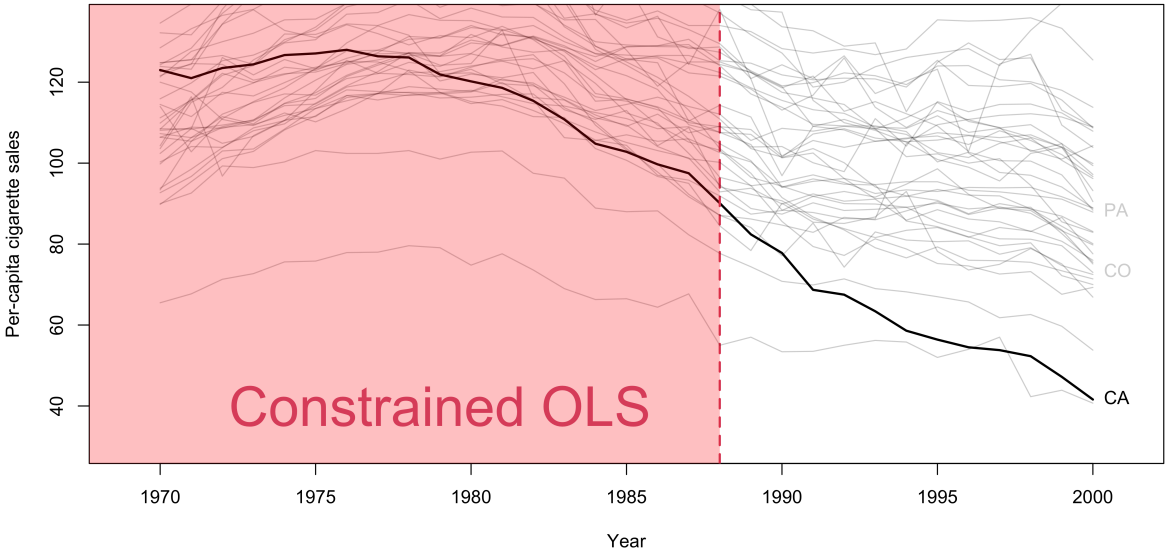

Per-capita cigarette sales
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$Y_t^{(1)}$
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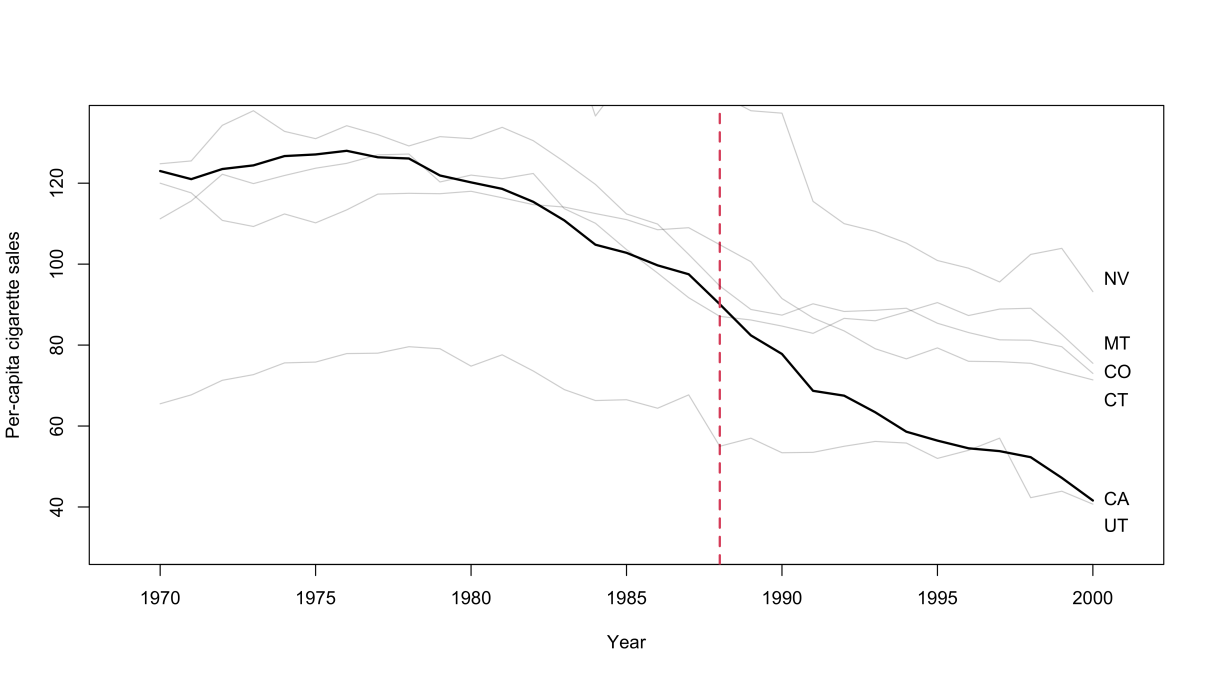

Per-capita cigarette sales
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τ_t^*
$Y_t^{(1)}$
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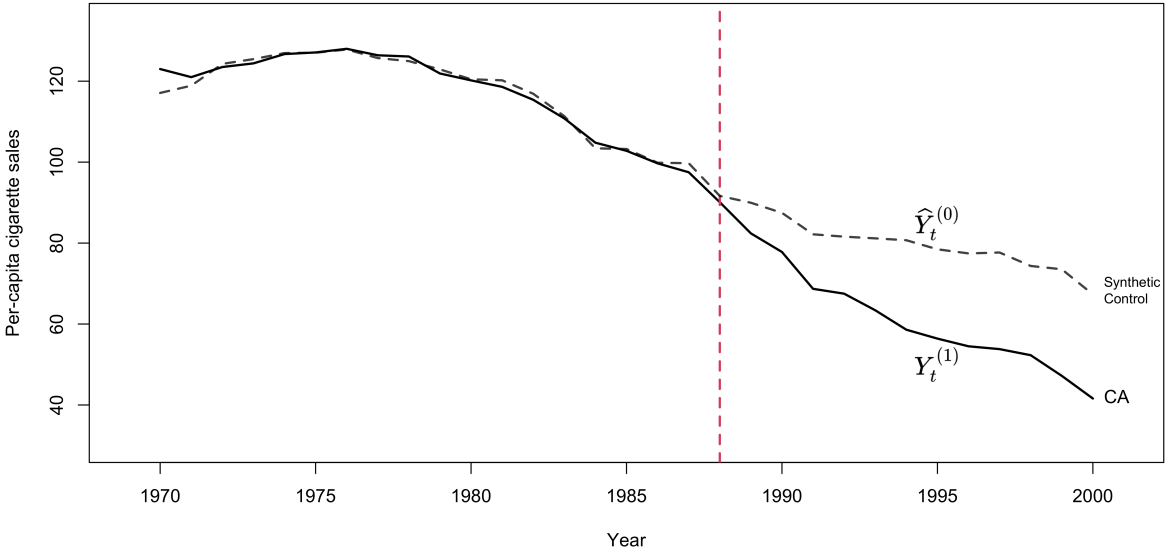

Per-capita cigarette sales
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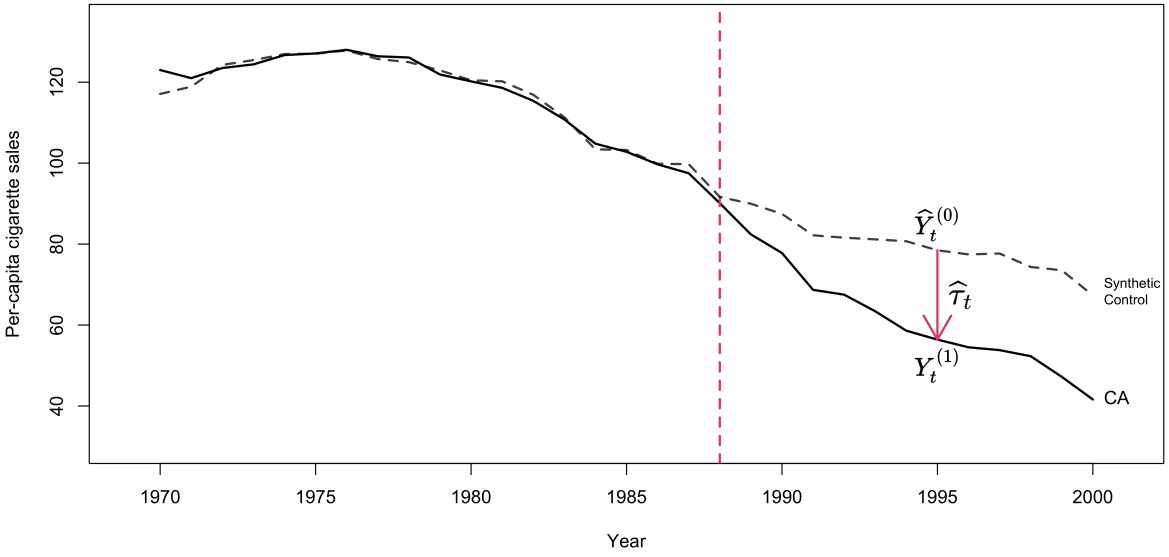

Per-capita cigarette sales
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Per-capita cigarette sales
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Per-capita cigarette sales
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Per-capita cigarette sales
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Per-capita cigarette sales
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1995
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Year
$\widehat{Y}_t^{(0)}$
Synthetic Control
$Y_t^{(1)}$
CA


Per-capita cigarette sales
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2000
Year
$\widehat{Y}_t^{(0)}$
$\widehat{\tau}_t$
$Y_t^{(1)}$
Synthetic
Control
CA

Synthetic Control Approach

- California tobacco control program in Abadie et al. (2010)

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- Estimand: Effect of California's Proposition 99 on California's per-capita cigarette sales

$$\tau_t = \mathrm{E}\big\{Y_t(1) - Y_t(0)\big\}\ , \quad 1988 < t$$

- Synthetic control: find a linear combination of other states

$$\widehat{Y}_t(0) \leftarrow \texttt{constrained OLS}(Y_t \sim W_t, \texttt{data} = \texttt{pre-trt data}) \quad \Rightarrow \quad \widehat{\tau}_t = Y_t - \widehat{Y}_t(0)$$

Formalizing the SC

Assumption (Synthetic Control)

(SUTVA) $Y_t = Y_t(A_t); \quad W_{it} = W_{it}(A_t)$

(No Interference) $W_{it}(1) = W_{it}(0)$

(Synthetic Control) *There exists a vector* $\gamma = (\gamma_1, \ldots, \gamma_N)^\intercal$ *such that* $\mathrm{E}\{Y_t(0) - \sum_{i=1}^N \gamma_i W_{it}\} = 0$ *for all* t

- No interference: the treatment assigned to unit $i = 0$ affects only that unit and has no impact on the outcomes of other units $i = 1, \ldots, N$

- Then, the treatment effect over the post-treatment period is identified:

$$\text{For } t = T_0 + 1, \ldots, T, \quad \mathrm{E}\{Y_t(1) - Y_t(0)\} = \mathrm{E}\{Y_t - Y_t(0)\} = \mathrm{E}\left(Y_t - \sum_{i=1}^N \gamma_i W_{it}\right)$$

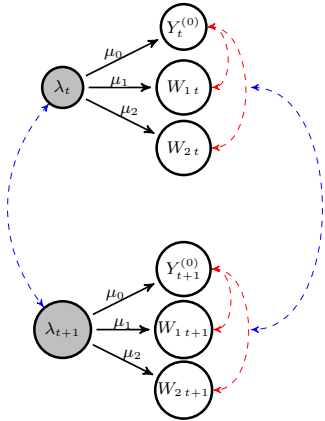
Interactive Fixed Effects Model

- Consider an IFEM (Bai, 2009)

$$\begin{aligned} Y_t(0) &= \mu_0^\intercal \lambda_t + \epsilon_{0t} \\ W_{it} &= \mu_i^\intercal \lambda_t + \epsilon_{it} \end{aligned}$$

λ_t: latent factors $\qquad$ μ_i: factor loadings

ϵ_{it}: error term with $\mathrm{E}(\epsilon_{it} \,|\, \lambda_t) = 0$



- Suppose there exists γ such that $\mu_0 = \sum_i \gamma_i \mu_i$

- Then, the synthetic control model is satisfied

$$\begin{aligned} \mathrm{E}[Y_t(0) - \sum_i \gamma_i W_{it}] &= \mathrm{E}[\underbrace{\left(\mu_0 - \textstyle\sum_i \gamma_i \mu_i\right)}_{=0}{}^\intercal \lambda_t + \underbrace{\epsilon_{0t} - \textstyle\sum_i \gamma_i \epsilon_{it}}_{\text{error}}] \\ &= 0 \end{aligned}$$

Estimation of SC

- IFEM with a SC

$$\begin{array}{rcccl} Y_t^{(0)} & = & \mu_0^\intercal \lambda_t & + & \epsilon_{0t} \\ W_{it} & = & \mu_i^\intercal \lambda_t & + & \epsilon_{it} \end{array} \quad , \quad \mu_0 = \sum_i \gamma_i \mu_i$$

- A linear model over the pre-treatment periods

$$Y_t = \sum_i \gamma_i W_{it} + \underbrace{\left(\epsilon_{0t} - \sum_i \gamma_i \epsilon_{it} \right)}_{\text{error} = \eta_t} \quad \Rightarrow \quad Y_t = \sum_i \gamma_i W_{it} + \eta_t$$

- Variants of OLS: Let $\gamma = (\gamma_1, \ldots, \gamma_N)^\intercal$ and $W_t = (W_{1t}, \ldots, W_{Nt})^\intercal$

OLS : $\widehat{\gamma}_{\text{OLS}} = \arg\min_\gamma T_0^{-1} \sum_{t=1}^{T_0} \left(Y_t - \gamma^\intercal W_t \right)^2$

Abadie et al. (2010) : $\widehat{\gamma}_{\text{Simplex}} = \arg\min_\gamma T_0^{-1} \sum_{t=1}^{T_0} \left(Y_t - \gamma^\intercal W_t \right)^2$ subject to $\|\gamma\|_1 = 1 \ , \ \gamma \geqslant 0$

Amjad et al. (2018) : $\widehat{\gamma}_{\text{Ridge}} = \arg\min_\gamma T_0^{-1} \sum_{t=1}^{T_0} \left(Y_t - \gamma^\intercal W_t \right)^2$ subject to $\|\gamma\|_2^2 \leqslant R$

Generally : $\widehat{\gamma}_{\mathcal{C}} = \arg\min_\gamma T_0^{-1} \sum_{t=1}^{T_0} \left(Y_t - \gamma^\intercal W_t \right)^2$ subject to $\gamma \in \mathcal{C}$

Estimation of SC

- A linear model over the pre-treatment periods

$$Y_t = \sum_i \gamma_i W_{it} + \underbrace{\left(\epsilon_{0t} - \sum_i \gamma_i \epsilon_{it} \right)}_{\text{error} = \eta_t} \quad \Rightarrow \quad Y_t = \sum_i \gamma_i W_{it} + \eta_t$$

- The OLS estimator is **inconsistent** because W_{it} and η_t are correlated through ϵ_{it}

- Ferman and Pinto (2021): the OLS estimator is consistent when either
  - $\epsilon_{it} = 0$ for all (i, t), i.e., noise-less setting; or
  - the true weight $= \arg\min_{\gamma \in \mathcal{C}} \|\gamma\|_2^2$

- These conditions are rather restrictive and not often met in practice
  - the noise-less setting ($\epsilon_{it} = 0$) is unrealistic in empirical applications;
  - the minimum-norm condition on the true weights is also unlikely

Inference of SC

- Inference in SC is challenging because the i.i.d. assumption is implausible; specifically:
  - The data are time series, but T is typically not large enough to rely on asymptotics
  - Only one unit receives treatment, making treatment assignment non-random

- Abadie et al. (2010): design-based permutation-based inference

- Li (2020): large-sample approximation

- Shaikh and Toulis (2021): cross-sectional permutation-based inference

- Ben-Michael et al. (2021); Cattaneo et al. (2021); Chernozhukov et al. (2021): build prediction intervals (conformal inference, non-asymptotic finite-sample guarantees)

Assessment of SC

- The SC estimator should have a good pre-treatment match:

$$Y_t \simeq \widehat{\gamma}^{\intercal} W_t \ , \qquad t \in \{\text{pre-trt}\}$$

- One can implement a placebo synthetic control by using only the pre-treatment period:
  - Suppose a placebo intervention is set at time P_0
  - Split the pre-treatment period into

$$\mathcal{T}_0 = \{1, \ldots, P_0\} \quad \text{and} \quad \mathcal{T}_1 = \{P_0 + 1, \ldots, T_0\}.$$

  - Fit a synthetic control estimator using $\mathcal{T}_0$ as the pre-treatment period and $\mathcal{T}_1$ as the post-treatment period
  - Since this is a fake intervention, the synthetic control estimator should ideally indicate a null effect over $\mathcal{T}_1$

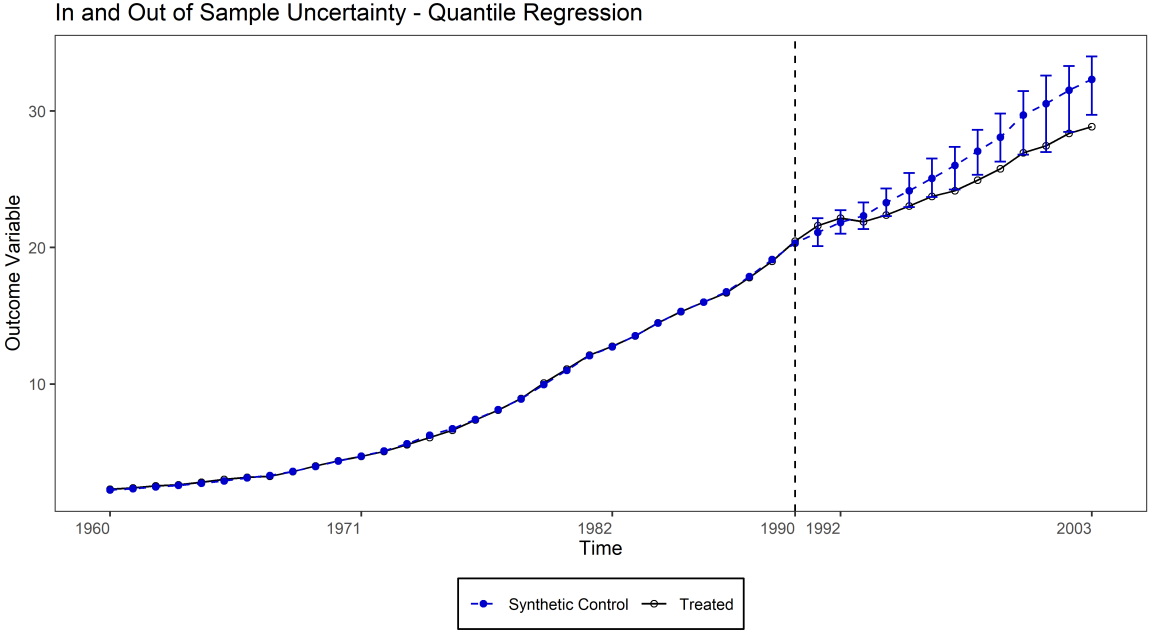
DiD vs. SC

	DiD	SC
Units	Many treated units Many control units	One treated units A few control units
Time period	Applicable for two-time point setting	Requires long pre-treatment period
Estimand	ATT (expected effect among trt units)	ATT (expected effect of the one trt unit)
Assumption	Parallel trends	Existence of SC
Inference	i.i.d. asymptotics ($N \to \infty$) Bootstrap	Time series asymptotics ($T \to \infty$) Prediction intervals

Example: German Reunification

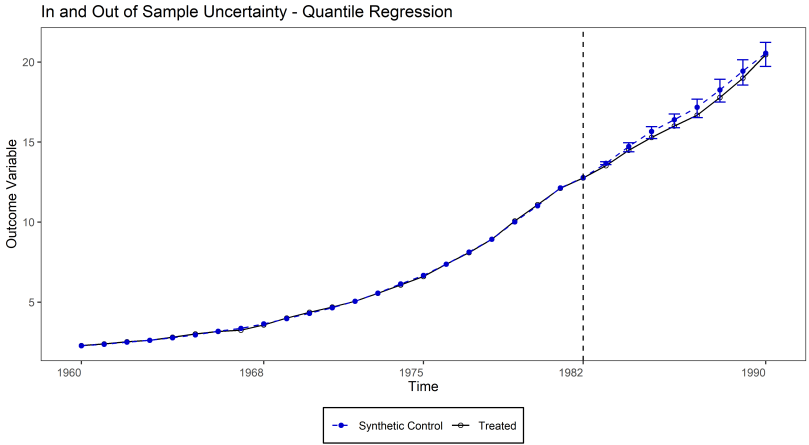
- Estimating the Economic Impact of German Reunification in 1989-1991
- $Y =$ West Germany's GDP per Capita
- $W =$ Other 16 OECD countries
- $t =$ 1960-2003

```
library(scpi)
data <- scpi_germany
df <- scdata(df = data, id.var = "country", time.var = "year",
             outcome.var = "gdp", period.pre = (1960:1990),
             period.post = (1991:2003), unit.tr = "West Germany",
             unit.co = setdiff(unique(data$country), "West Germany"))
result <- scpi(df)
scplot(result)
```


In and Out of Sample Uncertainty - Quantile Regression
Outcome Variable
30
20
10
1960
1971
1982
1990
1992
2003
Time
Synthetic Control
Treated

Example: German Reunification

```
df.placebo <- scdata(df = data, id.var = "country", time.var = "year",
             outcome.var = "gdp", period.pre = (1960:1982),
             period.post = (1983:1990), unit.tr = "West Germany",
             unit.co = setdiff(unique(data$country), "West Germany"))
result.placebo <- scpi(df.placebo); scplot(result.placebo)
```



Outline

1. Overview
2. Difference-in-Differences/Two-way Fixed Effects Models
3. Synthetic Control
4. Regression Discontinuity
5. Additional Methods
6. Assignment

Overview

- See the following material for comprehensive reviews
  - Cattaneo et al. (2024)
  - Causal Inference: The Mixtape

- Regression Discontinuity Design (RD or RDD):
  - Treatment assignment is determined by whether a unit crosses a threshold on a running variable (also referred to as assignment/forcing variable)
  - Example: A credit card is issued $(A = 1)$ if an applicant's credit score (running variable) exceeds a specified cutoff
  - The key assumption is that other factors do not change discontinuously at the threshold
  - Thus, any discontinuous change in the outcome at the cutoff can be attributed to the treatment

Example: The Effect of a College Degree from "Elite" Universities on Income

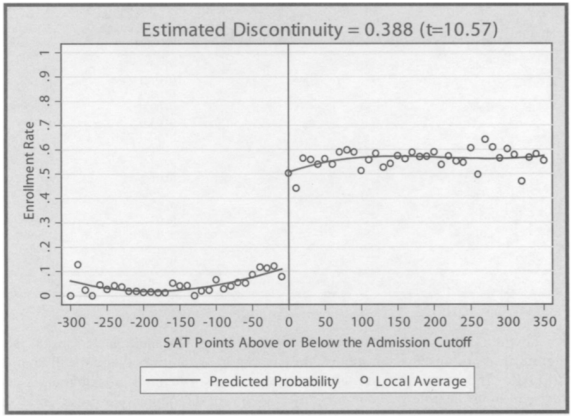
- Common conception: Individuals attending selective programs or schools earn higher incomes than those from less selective institutions

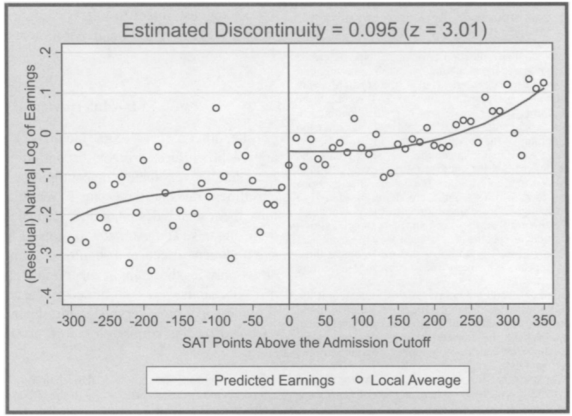
- Hard to know if higher income is due to:
  - causal effect: skills gained in the program actually raise income; or
  - selection bias: high-ability students are more likely to enter and would earn more regardless

- RD strategy: Compare the income of students with SAT scores slightly above the cutoff to those slightly below
  - Because their abilities are similar, any income difference can be attributed to degrees from selective programs

- Hoekstra (2009): effect attending the flagship state university (e.g., UIUC) on earning

Example: The Effect of a College Degree from "Elite" Universities on Income





RD Setting

- Y_i: Observed outcome of unit i
- A_i: 1 if unit i receives treatment; 0 otherwise
- R_i: Running variable

- In the previous example:
  - Y_i: Income
  - A_i: Attendance at the flagship state university
  - R_i: SAT score

RD Setting

- Y_i: Observed outcome of unit i
- A_i: 1 if unit i receives treatment; 0 otherwise
- $Y_i(a)$: Potential outcome of unit i under $A_i = a$
- R_i: Running variable of unit i

- In the previous example:
  - Y_i: Income
  - A_i: Attendance at the flagship state university
  - $Y_i(a)$: Income if $A_i = a$
  - R_i: SAT score

Two Types of RD

- Sharp RD:
  - The treatment and the running variable have a deterministic relationship:

$$A = \mathbb{1}(R \geqslant c)$$

  - In other words, the probability of receiving the treatment jumps at $R = c$ from 0 to 1

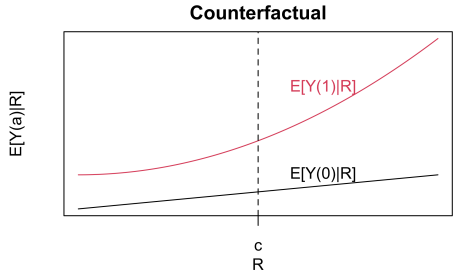
- Fuzzy RD:
  - the treatment and the running variable does not have a deterministic relationship
  - But, the probability of receiving the treatment jumps at $R = c$

Identification for Sharp RD

- Suppose $\mathrm{E}\{Y(a)|R=r\}$ is continuous at $r=c$ for $a=0,1$
  - Potential outcomes do not dramatically change at cutoff

- The treatment effect at $R=c$ is

$$\begin{aligned}\tau_c &= \mathrm{E}\{Y(1)-Y(0)|R=c\} \\ &= \mathrm{E}\{Y(1)|R=c\} - \mathrm{E}\{Y(0)|R=c\} \\ &= \lim_{\epsilon\downarrow 0}\mathrm{E}\{Y(1)|R=c+\epsilon\} - \lim_{\epsilon\downarrow 0}\mathrm{E}\{Y(0)|R=c-\epsilon\} \\ &= \lim_{\epsilon\downarrow 0}\mathrm{E}(Y|R=c+\epsilon) - \lim_{\epsilon\downarrow 0}\mathrm{E}(Y|R=c-\epsilon)\end{aligned}$$



Estimation for Sharp RD: Global Method

- A simple linear RD regression:

$$Y = \beta_0 + \beta_c A + \beta_r R + \epsilon$$

where $A = \mathbb{1}(R \geqslant c)$

  - This specification assumes the regression line exhibits a discontinuous "jump" at $R = c$, but maintains a linear trend with the same slope for all values of R

- A simple linear RD regression with varying slope:

$$Y = \beta_0 + \beta_c A + \beta_{r1} R + \beta_{r2}(R - c)_+ + \epsilon$$

where $(v)_+ = \max(v, 0)$

  - This specification allows the regression line to exhibit both a jump and a change in slope at $R = c$

- Polynomial regression:

  - Fit two separate polynomial regression models $Y \sim R$ over $\{R < c\}$ and $\{R \geqslant c\}$ groups

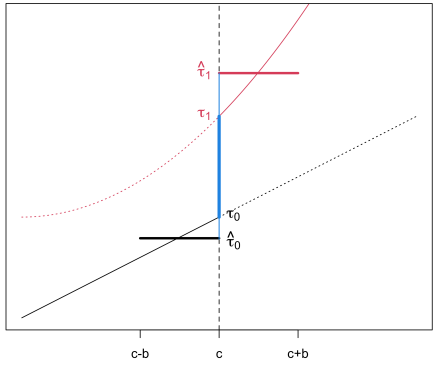
Estimation for Sharp RD: Local Method

- Local constant regression

$$\widehat{\tau}_1 = \frac{\sum_i \mathbb{1}(c \leqslant R_i < c+b) Y_i}{\sum_i \mathbb{1}(c \leqslant R_i < c+b)} \quad \text{vs} \quad \widehat{\tau}_0 = \frac{\sum_i \mathbb{1}(c-b < R_i < c) Y_i}{\sum_i \mathbb{1}(c-b < R_i < c)}$$

$b =$ bandwidth parameter

- $\widehat{\tau}_1 - \widehat{\tau}_0$ overstates $\tau_1 - \tau_0$ at the cutoff point
- there is a systematic bias in local constant regression estimates of the treatment effect



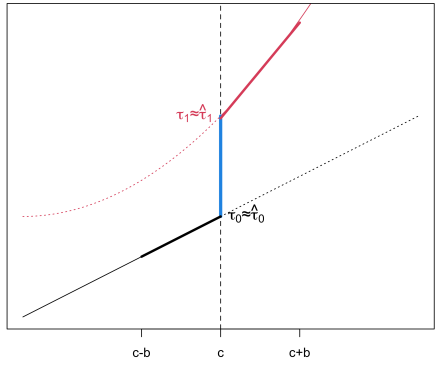
Estimation for Sharp RD: Local Method

- Local linear regression:

$$Y = \beta_0 + \beta_c A + \beta_{r1} R + \beta_{r2}(R-c)_+ + \epsilon$$

based on observations with $c - b < R < c + b$

- Hahn et al. (2001):
  - The bias from the local constant RD is substantially large comparable to the usual bias in kernel regression
  - The bias from the local linear RD is comparable to the usual bias in kernel regression



- Choice of bandwidth:
  - Hahn et al. (2001): Optimal $b \propto N^{-1/5}$
  - Fan and Gijbels (1992): Rule-of-thumb choice
  - Cross-validation (e.g., leave-one-out CV)

Identification for Fuzzy RD

- Treatment assignment:

$$\Pr(A = 1 \,|\, R) = \begin{cases} p_1(R) & \text{if } R \geqslant c \\ p_0(R) & \text{if } R < c \end{cases}$$

- One can consider $Z = \mathbb{1}(R \geqslant C)$ as an IV

- In the Sharp RD:

$$Y = \beta_0 + \beta_c A + f_y(R - c; \beta_f) + A \cdot g_y(R - c; \beta_g) + \epsilon$$

- The first stage of the Fuzzy RD:

$$A = \gamma_0 + \gamma_1 Z + f_a(R - c; \gamma_f) + Z \cdot g_a(R - c; \gamma_g) + \eta$$

- β_c is identified by the Wald-like statistics:

$$\beta_c = \frac{\lim_{\epsilon \downarrow 0} \mathrm{E}(Y|R = c + \epsilon) - \lim_{\epsilon \downarrow 0} \mathrm{E}(Y|R = c - \epsilon)}{\lim_{\epsilon \downarrow 0} \mathrm{E}(A|R = c + \epsilon) - \lim_{\epsilon \downarrow 0} \mathrm{E}(A|R = c - \epsilon)}$$

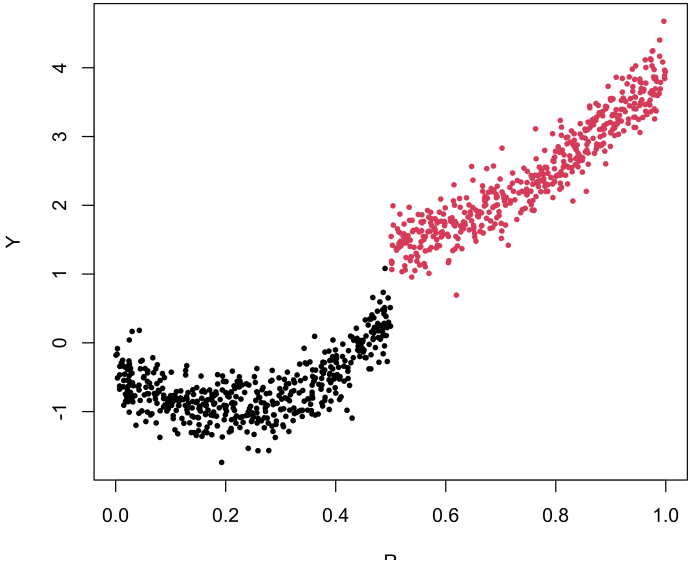
Estimation for Fuzzy RD

- Parametric (global) methods
- Nonparametric (local) methods

Example: Simulated Dataset

```
N <- 1000
R <- runif(N)
A <- as.numeric(R>0.5)
Y1 <- 3*R^3+1+rnorm(N)*0.25
Y0 <- 15*(R-0.2)^2-1+rnorm(N)*0.25
Y <- A*Y1+(1-A)*Y0
plot(R,Y,pch=19,col=A+1,cex=0.5)
```

- Effect at $c = 0.5 = 1.025$

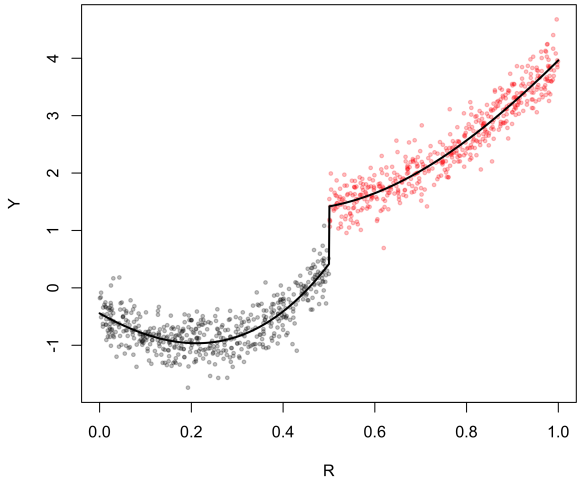


Example: Simulated Dataset

```
####### Parametric
Rc <- (R-0.5)
RD.Para <- lm(Y~1+A+Rc+I(Rc^2)+I(Rc^3)+
                A:Rc+A:I(Rc^2)+A:I(Rc^3))
summary(RD.Para)

R.Test <- seq(0,1,length=1001)
Rc.Test <- R.Test-0.5
NewData <- cbind(1,as.numeric(R.Test>0.5),
                 Rc.Test,Rc.Test^2,Rc.Test^3,
                 as.numeric(R.Test>0.5)*
                   cbind(Rc.Test,Rc.Test^2,Rc.Test^3))

plot(R,Y,pch=19,
     col=ifelse(A==0,rgb(0,0,0,0.25),rgb(1,0,0,0.25)),
     cex=0.5)
points(R.Test,NewData%*%coef(RD.Para),type='l',lwd=2)
```



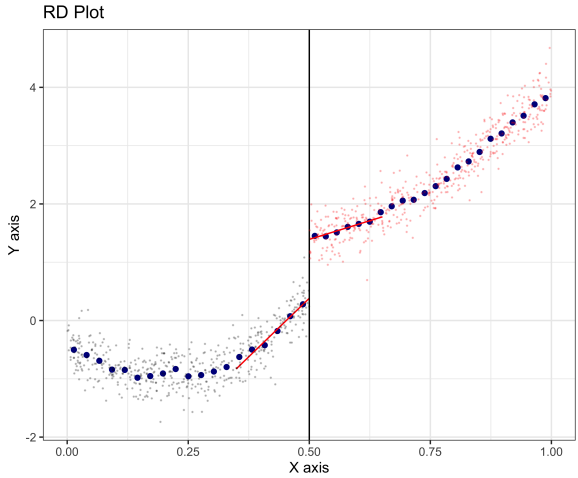
- $\widehat{\tau}_c = 1.003$, SE $= 0.066$, 95% CI $= (0.876, 1.130)$

Example: Simulated Dataset

```
####### Local Linear
library(rdrobust)
RD.LL <- rdrobust(y=Y,x=R,c=0.5)
summary(RD.LL)

RD.LL.Plot <- rdplot(y = Y, x = R, c = 0.5,
                     binselect = "es",
                     kernel = tolower(RD.LL$kernel),
                     p=RD.LL$p, h=RD.LL$bws[1],
                     x.lim = c(0,1), ci=95)

new_points <- data.frame(x=R,y=Y,a=A)
RD.LL.Plot$rdplot +
  geom_point(data = new_points,
             aes(x = x, y = y,
                 color = ifelse(a == 0,
                                rgb(0,0,0,0.2),
                                rgb(1,0,0,0.2))),
             size = 0.15) +
  scale_color_identity()
```



- $\widehat{\tau}_c = 1.010$, SE $= 0.073$, 95% CI $= (0.842, 1.119)$

Outline

1. Overview
2. Difference-in-Differences/Two-way Fixed Effects Models
3. Synthetic Control
4. Regression Discontinuity
5. Additional Methods
6. Assignment

Additional Popular Methods in Social Sciences

- Changes-in-changes, Quantile DiD (Athey and Imbens, 2006)
- Nonlinear DiD (Puhani, 2012; Wooldridge, 2022)
- Distributional DiD (Bonhomme and Sauder, 2011; Fan and Yu, 2012; Callaway et al., 2018; Callaway and Li, 2019; Tchetgen Tchetgen et al., 2024)
- Synthetic DiD (Arkhangelsky et al., 2021)
- Matrix completion methods (Athey et al., 2021)

Outline

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Assignment: DiD

- Analyze the Zika virus dataset available from:

  `read.csv("https://raw.githubusercontent.com/qkrcks0218/UDiDEpi/refs/heads/main/Zika_Brazil.csv")`

- Use the following variables
  - X = `X_1,X_2,X_3` = log population, population density, proportion of females in 2014
  - A = `Trt` = Pernambuco (PE)
  - Y = `Y` = Birth rate
  - t = `time` = 0/1 = 2014/2016

- Goal: Estimate the causal effect of the Zika virus outbreak in birth rate

- Estimate the effect based on the DiD method and TWFE model

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

Assignment: SC

- Analyze the California tobacco data available from:

```
read.csv("https://raw.githubusercontent.com/synth-inference/synthdid/refs/heads/master/data/california_prop99.csv", sep=";")
```

- Use the following variables
  - Y = California's `PacksPerCapita`
  - W = Other states' `PacksPerCapita`
  - T_0 = 1988 (California's Proposition 99 was implemented in 1989)

- Goal: Effect of California's Proposition 99 on California's per-capita cigarette sales

- Estimate the effect using the SC method

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

Assignment: RD

- Analyze the dataset analyzed in Lee (2008) available from

```
library(RDHonest); lee08
```

- Use the following variables
  - $R =$ `margin` $=$ Democratic margin of victory in the previous election
  - $Y =$ `vote` $=$ Vote share in next election
  - $c = 0$

- Goal: Effect of incumbency advantage, i.e., how winning an election influences a candidate's vote share in the subsequent election

- Estimate the effect using the RD method based on:
  - parametric approach (e.g., linear regression)
  - nonparametric approach (e.g., local linear regression)

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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